

Indian Institute of Technology Kharagpur
Department of Mathematics
MA11003 - Advanced Calculus
Problem Sheet -13B

1. (a) If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are constant vectors, then show that $\vec{r} = \vec{a}t^2 + \vec{b}t + \vec{c}$ is the path of a particle moving with constant acceleration.
(b) Prove that a non-constant vector $\vec{u}$ has a constant length if and only if $\vec{u} \cdot \frac{d\vec{u}}{dt} = 0$.
2. Evaluate the following limits
 - (a) $\lim_{t \rightarrow 1} \vec{r}(t)$, where $\vec{r}(t) = e^{t-1}\hat{i} + 4t\hat{j} + \frac{t-1}{t^2-1}\hat{k}$.
 - (b) $\lim_{t \rightarrow 2} \vec{r}(t)$, where $\vec{r}(t) = \frac{1-e^{t+2}}{t^2+t+2}\hat{i} + \hat{j} + (t^2+6t)\hat{k}$.
 - (c) $\lim_{t \rightarrow 1} \vec{r}(t)$, where $\vec{r}(t) = t^3\hat{i} + \frac{\sin(3t-3)}{t-1}\hat{j} + e^{2t}\hat{k}$.
3. Determine the vector equation for the line segment that starts at the point $P = (x_1, y_1, z_1)$ and ends at the point $Q = (x_2, y_2, z_2)$.
4. Find the gradient and the unit normal vector to the following surfaces
 - (a) $x^2 + y - z = 4$ at the point $(2, 0, 0)$.
 - (b) $x^2 + 2y^2 + 3z^2 = 0$ at the point $(\sqrt{10}, 0, 0)$.
 - (c) $x^2y + 2xz = 4$ at the point $(2, -2, 3)$.
5. Find the directional derivatives of the following scalar valued functions
 - (a) $f(x, y) = e^x \cos y$ at the point $(0, \frac{\pi}{4})$ in the direction of $(\hat{i} + 3\hat{j})/\sqrt{10}$.
 - (b) $f(x, y, z) = e^x + yz$ at the point $(1, 1, 1)$ in the direction of $\hat{i} - \hat{j} + \hat{k}$.
 - (c) $f(x, y, z) = \frac{1}{x^2+y^2+z^2}$ at the point $(2, 3, 1)$ in the direction of $\hat{i} + \hat{j} - 2\hat{k}$.
6. Find the directional derivative of the scalar valued function $f(x, y) = \frac{y}{x^2 + y^2}$ at the point $(0, 1)$ in the direction of a vector which makes an angle of 30° with the positive x -axis.
7. (a) In what direction from the point $(1, 3, 2)$ the directional derivatives of $\phi = 2xz - y^2$ is maximum? What is the magnitude of this maximum?
(b) Find the values of the constant a , b and c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^3$ at the point $(1, 2, -1)$ has maximum of magnitude 64 in the direction of the z -axis.
8. If $r = |\vec{r}|$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then prove that
 - (a) $\nabla\left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$.
 - (b) $\nabla(\log(|\vec{r}|)) = \frac{\vec{r}}{r^2}$.

- (c) $\nabla(r^n) = nr^{n-2}\vec{r}$.
9. Let $\vec{F} = 2xz^2\hat{i} + \hat{j} + xy^3z\hat{k}$ and $f = x^2y$. Then compute the following
- $\text{curl}(\vec{F})$
 - $\vec{F} \times \nabla f$
 - $\vec{F} \cdot (\nabla f)$
10. For any two vector fields $\vec{F}$ and $\vec{G}$ show that
- $\nabla \cdot (\nabla \times \vec{F}) = 0$
 - $\text{div}(\vec{F} \times \vec{G}) = \text{curl}(\vec{F}) \cdot \vec{G} - \text{curl}(\vec{G}) \cdot \vec{F}$
 - $\nabla \times (\nabla \vec{F}) = \vec{0}$
11. For all smooth scalar field f and g , show that
- $\nabla(fg) = g\nabla f + f\nabla g$
 - $\nabla\left(\frac{f}{g}\right) = \frac{1}{g^2}(g\nabla f - f\nabla g)$
12. Check whether $\vec{F}$ is a conservative vector field or not. If it is, then find the corresponding potential function, where
- $\vec{F} = (2xy, x^2 + 2yz, y^2)$
 - $\vec{F} = (2xy + z^3, x^2, 3xz^2)$
13. (a) Find the values of the constant a , b and c so that the vector $\vec{w} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ becomes irrotational.
 (b) Determine the constant a so that the vector $\vec{v} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$ is solenoidal.
14. Let $\vec{F} = yz^2\hat{i} + xy\hat{j} + yz\hat{k}$ be a vector field. Then what is the value of $\text{div}(\text{curl}(\vec{F}))$?
15. Let $\phi(x, y, z) = x^2y - xez$, $P_0 = (2, -1, \pi)$ and $\vec{u} = \frac{1}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k})$. Then what is the rate of changes of $\phi(x, y, z)$ at the point P_0 in the direction of the vector $\vec{u}$?
16. Consider the surface $\phi(x, y, z) = z - \sqrt{x^2 + y^2} = 0$. Then find the normal vector and the tangent plane to the surface at the point $(1, 1, \sqrt{2})$.
-